Arrays, selectors and three-dimensional integration

A single RRAM cell must become part of an addressable network. This chapter follows the current through that network, then examines access devices, sensing and the physical layout.

All numerical examples are deliberately simplified circuit models, not measured device specifications.

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A crosspoint is a cell, not an isolation switch

At the intersection of row R_i and column C_j, a memory element connects the two lines. Applying a voltage to a selected row and sensing a column also establishes voltages across other intersections.

Those other cells can carry currents into the sense circuit, or form a path through floating lines. Which currents appear depends on the line-bias and sensing circuit; a cell's HRS/LRS ratio alone does not determine array performance.

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Fixed $V/2$ bias: define every line

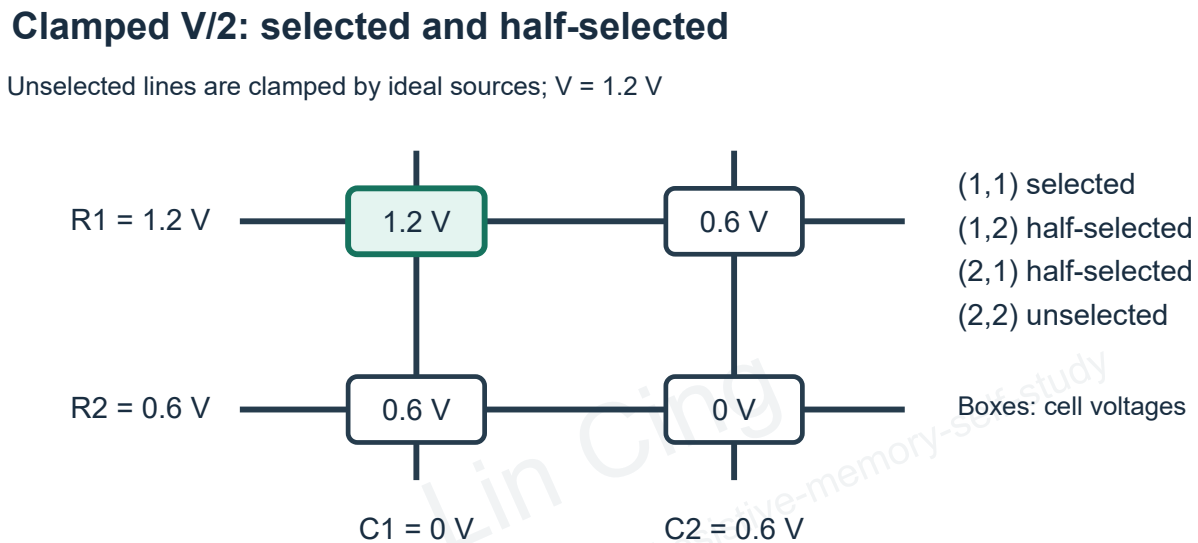
Drive the selected row to V, the selected column to 0 and every unselected row and column to $V/2$. Assume ideal voltage sources and wires.

Cell position	Row potential	Column potential	Row − column
Selected intersection	V	0	V
Selected row only	V	$V/2$	$V/2$
Selected column only	$V/2$	0	$V/2$
Neither selected	$V/2$	$V/2$	0

For $V = 1.2$ V the half-selected cells see 0.6 V. This is one bias scheme; finite line resistance or voltage division within a 1S1R cell changes the device-level voltages.

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Fixed-bias example: follow the sources



The unselected lines are **clamped**, not floating: their sources can supply or absorb current. The selected-column current can contain both the selected-cell current and current from a half-selected cell.

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Half-select is a stress condition

An unselected cell experiences a fraction of the applied pulse. Whether it remains unchanged depends on its initial state, device voltage, polarity, pulse width and number of repetitions—not on voltage alone.

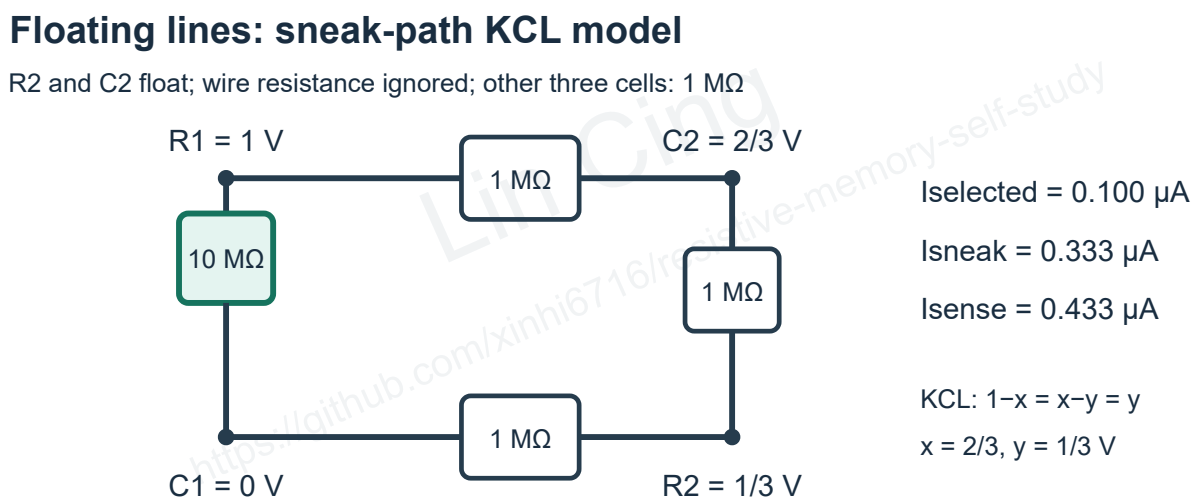
For an array, count the pulses imposed on each half-selected cell by its neighbors' writes. Compare its resulting distribution against the read criterion, including variability and cumulative disturb.

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Floating lines create a different network

Take a 2×2 passive array: row $R_1 = 1$ V and column $C_1 = 0$ V; R_2 and C_2 float. The selected cell R_1–C_1 has conductance $0.1\ \mu$S; each other cell has $1\ \mu$S.

Assume linear, fixed-state conductances, ideal wires and no connection from a floating line to an external load. A path $R_1 \to C_2 \to R_2 \to C_1$ now runs in parallel with the selected cell.



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Solve the floating network by KCL

Let the unknown potentials of C_2 and R_2 be x and y volts. With all three unselected conductances equal to $1\,\mu$S, Kirchhoff's current law gives

$$(x-1)+(x-y)=0, \qquad (y-x)+y=0.$$

Thus $2x-y=1$ and $2y=x$, so $x=2/3$ V and $y=1/3$ V. Each of the three series branches drops $1/3$ V.

Selected-cell current is $0.1\,\mu$A; the alternative path carries $1/3\,\mu$A. The sense node at C_1 receives $0.1+1/3=0.433\,\mu$A, approximately.

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Compare the boundary conditions

In fixed $V/2$ bias the other lines are held by sources; in the floating example their voltages are **unknowns determined by KCL**. The three-resistor path calculation applies to that floating 2×2 topology, not automatically to a clamped array.

An HRS selected cell surrounded by LRS cells is an illustrative adverse pattern: the unwanted current can exceed the selected current. The actual worst pattern depends on the complete bias, device and sensing model.

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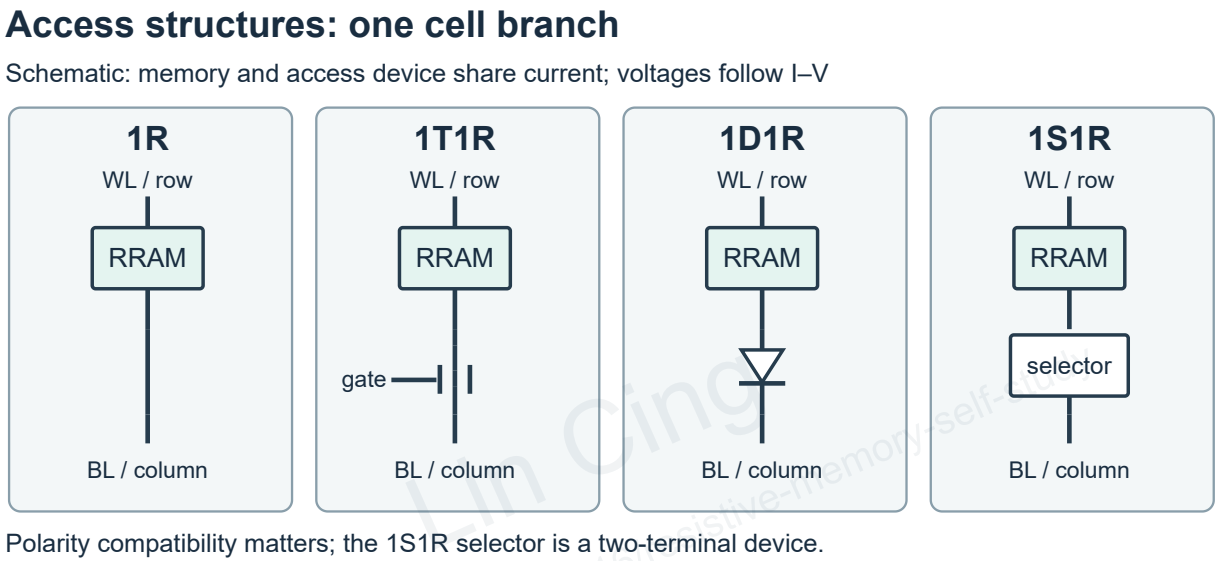
Access structures change the available controls

Structure	Access mechanism	Design consequence
1R	Memory element alone	Isolation relies on its own I–V and bias scheme.
1T1R	A gated transistor in series	Gate control can suppress leakage and limit current, at a footprint and voltage cost.
1S1R	Two-terminal selector in series	Selector nonlinearity suppresses some low-bias currents without a gate.
1D1R	Diode in series	Rectification favors one direction; compatibility with both programming polarities must be checked.

A self-selective element incorporates access nonlinearity into its stack; it does not provide the separate gate of 1T1R.

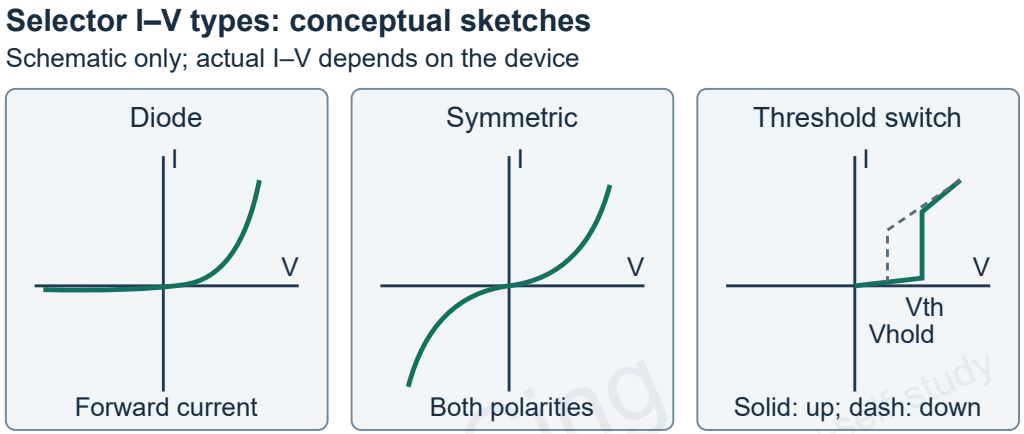
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Access structures: follow each control path



The transistor has a separate gate terminal. A two-terminal selector or diode is driven through the series path; the memory and its access element share the applied voltage.

Three selector response families



A **rectifier** conducts asymmetrically in the two polarities. A **varistor** gives a strongly nonlinear response; it may be engineered for conduction in both directions, but symmetry is not automatic. A **threshold switch** can enter a conducting state after a threshold and return when its holding condition is lost.

Each label describes a response family, not a guarantee of leakage, switching speed, retention or suitability for a given memory.

Threshold and holding conditions

For a hypothetical volatile selector, an increasing voltage reaches a turn-on threshold V_{th}; after turn-on, the current and device voltage follow the selector's conducting branch. The selector returns to its off branch when current falls below a holding current I_h in the stated model.

The conducting branch can have hysteresis, and real thresholds depend on polarity, pulse duration and history. A threshold selector is not an ideal permanent latch and must not be confused with the nonvolatile state stored in RRAM.

The source, wire and series memory constrain the trajectory: crossing an isolated-device threshold alone does not prove a stable operating point.

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A series selector shares voltage with memory

The same current flows through both devices and their voltages satisfy $V_{app} = V_S + V_M$. For nonlinear devices, solve $I_S(V_S) = I_M(V_M)$ together with that sum, using the relevant state, polarity and branch.

For a **fixed-state linear illustration**, let $R_S = 2\ \text{k}\Omega$, $R_M = 4\ \text{k}\Omega$ and $V_{app} = 1.2\ \text{V}$. Then $I = 1.2/6\,\text{k}\Omega = 0.2\ \text{mA}$; $V_S = 0.4\ \text{V}$ and $V_M = 0.8\ \text{V}$. Neither element receives the full applied voltage.

For a threshold selector use its actual I–V branches and circuit load line; fixed-resistance division cannot predict the switching transient or the holding condition.

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A nonlinear operating point: a piecewise model

Assume a positive-bias, non-hysteretic selector with $I = V_S/(100\,\mathrm{k\Omega})$ for $V_S < 0.8$ V, and $I = (V_S - 0.8\,\mathrm{V})/(1\,\mathrm{k\Omega}) + 8\,\mu$A for $V_S \geq 0.8$ V. A fixed $4\,\mathrm{k\Omega}$ memory is in series; $V_{app} = 1.2$ V.

Try the low-voltage branch: $I = 1.2\,\mathrm{V}/(104\,\mathrm{k\Omega}) = 11.54\,\mu$A and $V_S = 1.154$ V. That violates $V_S < 0.8$ V, so reject it.

On the high branch, $V_S = 0.8\,\mathrm{V} + (1\,\mathrm{k\Omega})(I - 8\,\mu\mathrm{A})$ and $1.2\,\mathrm{V} = V_S + (4\,\mathrm{k\Omega})I$. The intersection is $I = 81.6\,\mu$A, $V_S = 0.8736$ V and $V_M = 0.3264$ V, consistent with the branch. This is a teaching model, not a measured selector.

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Check the read and write windows separately

At the intended read voltage, distinguish HRS and LRS while limiting leakage and avoiding read disturb. At both SET and RESET polarities, the combined stack must deliver the memory's required voltage, current and pulse duration.

Also check selector on-current, half-select leakage, endurance, temperature, speed and variability. A selector that blocks reverse current may be suitable for a particular unipolar scheme but conflict with bipolar operation; topology alone cannot establish compatibility.

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What a nonlinearity ratio tells us

For a specified device state, polarity and measurement timing, define

$$NL = \frac{|I(V_{read})|}{|I(V_{read}/2)|}.$$

The denominator must be nonzero, and numerator and denominator must use the same current sign or magnitudes consistently. For $V_{read} = 1.2\,\mathrm{V}$, $NL = 10^3$ and $|I(1.2\,\mathrm{V})| = 10\,\mu$A, the model gives $|I(0.6\,\mathrm{V})| = 10$ nA.

This synthetic example compares only two points. It does not specify other voltages, switching dynamics, noise or complete-array read margin.

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Read margin needs a sensing definition

Let I_H and I_L denote the sensed currents for a selected HRS and LRS, including all paths admitted by the chosen bias circuit. In a simple current-threshold reader, choose T and call the state LRS if $I_{sense} > T$.

If the possible HRS currents lie in $[H_{min}, H_{max}]$ and LRS in $[L_{min}, L_{max}]$, a deterministic threshold separating both ranges exists only if $H_{max} < L_{min}$. The separation $L_{min} - H_{max}$ is a current-domain margin under this definition.

Changing the sense circuit or using a voltage-domain margin changes the calculation; noise and distributions must be included before claiming an error rate.

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A small current-margin calculation

Use the previous floating 2×2 network, assume the three unselected cells remain linear at $1\ \mu$S, and switch only the selected conductance between 0.1 and $1\ \mu$S. Both states see 1 V across the selected cell and have the same $1/3\ \mu$A alternative-path current.

Hence $I_H = 0.433\ \mu$A and $I_L = 1.333\ \mu$A. With threshold $T = 0.8\ \mu$A, the HRS and LRS separations from threshold are 0.367 and $0.533\ \mu$A respectively. These nominal values assume an ideal current sense at 0 V, no noise and no state variation.

If unselected states or node voltages vary between reads, recalculate both ranges. The example does not establish how margin scales for a larger array.

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Threshold window with an assumed background

For a separate example, hold the sense column at 0 V and assume the sneak contribution is fixed at $0.3\ \mu$A regardless of the selected state. Let the selected HRS and LRS currents be 0.1 and $1.0\ \mu$A, respectively. The sensed currents are then $I_H = 0.4\ \mu$A and $I_L = 1.3\ \mu$A.

With the rule “LRS if current is strictly greater than T,” both nominal states are classified correctly when $0.4 \le T < 1.3\ \mu$A. This definition gives a current window $I_L - I_H = 0.9\ \mu$A; other definitions may normalize the margin differently.

The $0.3\ \mu$A is an explicit simplifying assumption, not the $1/3\ \mu$A computed for the earlier floating network. In general solve the entire network separately for HRS and LRS.

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Add variability and bounded sensing error

In a particular operating condition, suppose measured HRS currents lie in $[0.35, 0.55]$ μA and LRS currents in $[1.10, 1.40]$ μA. Let the sensing error have absolute value at most 0.05 μA. The resulting observed ranges are $[0.30, 0.60]$ and $[1.05, 1.45]$ μA.

Using the same strict-greater-than threshold rule, every value in these bounds is classified correctly for $0.60 \leq T < 1.05$ μA. If the ranges overlap, no single threshold guarantees correct classification for all bounded cases.

This is a bounded-error example. Actual distributions, sample sizes, correlations and failure rates need separate evaluation.

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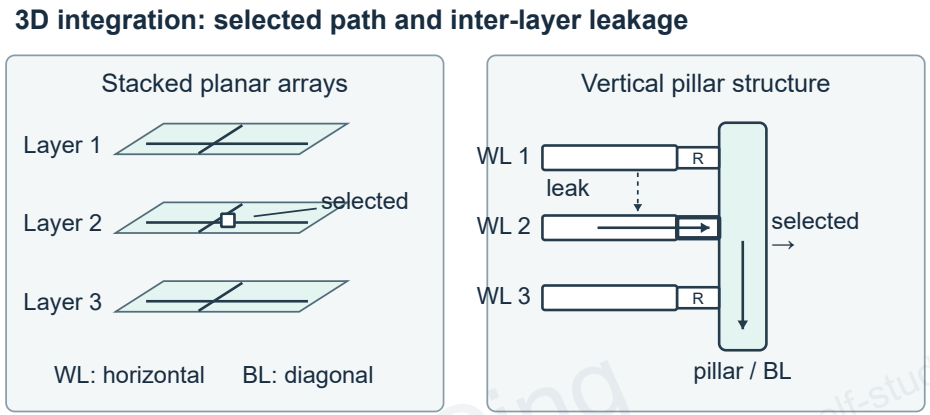
Array size: where leakage accumulates

Larger arrays generally introduce more possible paths and longer wires, but the actual sensed leakage depends on topology, biases and nonlinear I–V characteristics. A sum of identical half-select currents is valid only if the assumed circuit really presents those same device voltages and currents.

To evaluate a proposed size, solve the network for representative data patterns and process corners, then include sense noise, timing, supply and reference uncertainty. A single two-point NL cannot supply these missing conditions.

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Horizontal layers and vertical pillars



Leakage depends on structure and bias.

Horizontal stacking repeats lateral crossbar planes. In a vertical structure, memory cells occur where a pillar electrode meets separately addressable stacked word-line layers. The selected path and isolation depend on the particular layout and line bias.

Interlayer leakage, alignment, parasitic capacitance, contacts, heat and yield can offset the apparent density gain. Different 3D architectures cannot be ranked without a stated circuit and manufacturing comparison.

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Interpret an area calculation correctly

For an idealized planar crosspoint with pitch $2F$ in each direction, area per cell is $(2F)^2 = 4F^2$. If $F = 20$ nm, this is $1600\,\mathrm{nm}^2$.

With three perfect layers over the same projected footprint and one bit per cell, the ideal projected cell area per bit is $1600/3 \approx 533.3\,\mathrm{nm}^2$. This is an accounting estimate, not the area of a manufactured memory chip.

Peripheral decoders, sensing, routing, redundancy, layer isolation, spacing and yield must be included in a system comparison. Extra layers can change process cost and thermal constraints.

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Self-selecting memory: the read path is state-dependent

A self-selecting cell combines nonvolatile storage and a sufficiently nonlinear access response in the same two-terminal stack. A separate selector is not required **only when** the complete array meets its leakage, disturb and sensing specifications at the chosen size and biases. It is not an ideal off-switch or a guarantee of bipolar operation.

Characterize both HRS and LRS, both polarities and pulse histories. Compare selected-state currents at V_{read} and the corresponding half-selected currents at the actual device voltages. A single NL value from one state cannot specify the full network or its retention and endurance.

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A conditional leakage budget

Assume a sense column has an ideal 0 V sense node, one selected cell at V_{read}, and 99 other branches each held in the LRS at the same half-select voltage. Each LRS branch conducts 10 nA at half bias and $10\,\mu$A at full bias, giving $NL = 1000$ for this state. The selected HRS current at full bias is $0.2\,\mu$A.

If the 99 branch currents remain fixed when the target changes state, their total is 99×10 nA $= 0.99\,\mu$A. The sensed HRS and LRS currents are 1.19 and $10.99\,\mu$A. The background exceeds the selected HRS current, but that alone does not imply a read error: evaluate the state distributions, sensing errors and threshold margin. A nonlinearity ratio alone does not establish the number, voltages or states of the leakage paths.

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Oxygen profiles are engineered across the stack

For a **particular VCM stack**, one design separates a vacancy-rich, relatively oxygen-poor region from a more nearly stoichiometric switching or tunnelling region. Layers of different composition and thickness can shape where oxygen vacancies are supplied and where a conductive path ruptures. The local profile is a design hypothesis to test, not a universal equilibrium diagram.

Reactive electrodes such as Ti or Hf may exchange oxygen with an adjacent oxide; an opposite, less reactive contact can contribute to an asymmetric boundary. Electrode oxidation, diffusion through imperfect barriers and ambient exposure can alter that intended profile. The electrochemical potentials, contact reactions and manufacturing history must be specified before predicting a direction.

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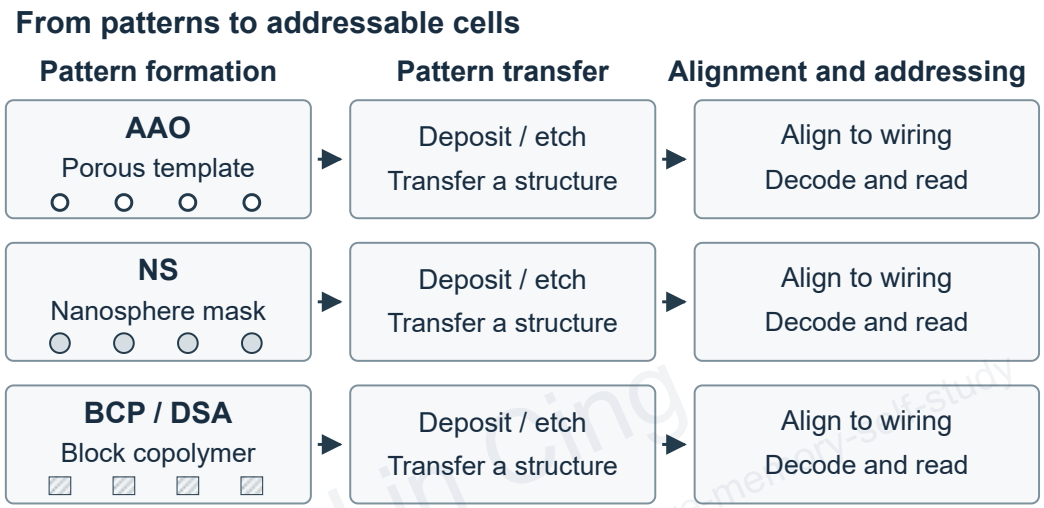
Dopants, alloys and buffer layers: what is controlled?

In VCM, dopants can alter vacancy formation energy, charge compensation and migration barriers. A material's response depends on dopant concentration, position, charge state and processing; the name of a dopant alone does not guarantee forming-free operation or improved retention.

In ECM, Cu or Ag from an active electrode supplies mobile ions. An alloy can change the chemical activity and release of the active species; a buffer layer can restrict unwanted diffusion or modify deposition geometry. Counter-electrode kinetics and electrolyte structure also affect the path. Test forming, switching variability, retention and disturbance together rather than optimizing only mobility.

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Three templates, one addressing problem



A regular pattern is not automatically addressable: contacts, alignment and defects matter.

AAO pores, nanosphere masks and guided block-copolymer domains form patterns by different processes. After transferring the pattern into the intended layer, registration to the electrode lines and continuity of each contact must be verified. A conductive-probe measurement at one site is useful evidence for that site; it is not by itself an electrically addressed memory array. A continuous switching layer may also create lateral leakage and crosstalk between cells.

Compare formation, transfer and registration

Method	Form a template	Transfer and registration
AAO	Anodize Al to generate pores; optionally open the pore bottoms.	Deposit into or through pores, or transfer the template. Guided initial patterning and alignment are separate engineering tasks.
Nanosphere lithography	Assemble a sphere monolayer.	Use spheres or their openings as a deposition/etch mask; remove the mask, inspect defects and align to contacts.
BCP / DSA	Anneal phase-separated polymer domains; a guide pattern constrains their location.	Selectively remove or convert one domain, transfer by etch or lift-off, then verify registration and isolation.

Template geometry, defect density, process compatibility and alignment depend on the actual process; these methods cannot be ranked from nominal feature size alone.

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Individual success versus a completely good array

Suppose 100 independently fabricated contacts each pass a specified electrical test with probability $p = 0.99$. With no repair or redundancy, the probability that **all** pass is

$$P(\text{all pass}) = 0.99^{100} \approx 0.366.$$

If X counts passing contacts, linearity of expectation gives $E[X] = 100(0.99) = 99$. This mean needs equal marginal pass probabilities but **not** independence; the all-pass product requires independence. Spatially correlated defects or repair change the whole-array yield, and a successful contact alone does not establish a functioning memory cell.

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A reproducible array comparison

State the stack, access structure, array dimensions, line biases, reference polarity, wire and sense models, pulses and timing. Record whether the reported current includes unselected paths and whether the result comes from measurements, simulation or extrapolation.

Report temperature, sample count, cycle history, variation, failure criterion and relevant uncertainty. A particular experimental selector or 3D stack is evidence for that configuration, not a universal specification for all RRAM arrays.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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